

Energetics

Exothermic reaction: A reaction that gives out heat energy to the surrounding, including mixture and air. It results in increasing the temperature of the surrounding.

Examples:

- 1- Burning / combustion
- 2- Rusting
- 3- Neutralization reactions
- 4- Displacement reactions
- 5- Freezing and condensing
- 6- Voltaic cell

Endothermic reaction: A reaction that takes in heat energy from the surroundings. It results in the drop of temperature of the surroundings.

Examples:

- 1- Photochemical reactions

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Photosynthesis

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Photography
- 2- The reaction between nitrogen & oxygen.
$$\text{N}_2 + \text{O}_2 \rightarrow 2\text{NO}$$
- 3- Thermal decomposition
- 4- Melting and boiling
- 5- Electrolysis.

Activation energy: E_a : The minimum energy that the colliding particles must have in order to react.

→ To start a reaction, bonds must be broken, and particles must first collide with each other. They must collide with sufficient energy → Activation energy.

- * For some reactions, the energy from surroundings is sufficient.
- * Some exothermic reactions need heat from a bunsen burner, or just a single spark to start bond breaking. The energy given out when new bonds form will break further bonds.
- * Endothermic reactions, have high activation energy, so heating must be continued until the reaction is complete.

Enthalpy: The energy content of a system at a constant pressure.

→ We cannot measure the enthalpy of a system, but we can measure the enthalpy change. If a system takes in or gives out energy, it undergoes enthalpy change.

$$\Delta H = H_{\text{products}} - H_{\text{reactants}}$$

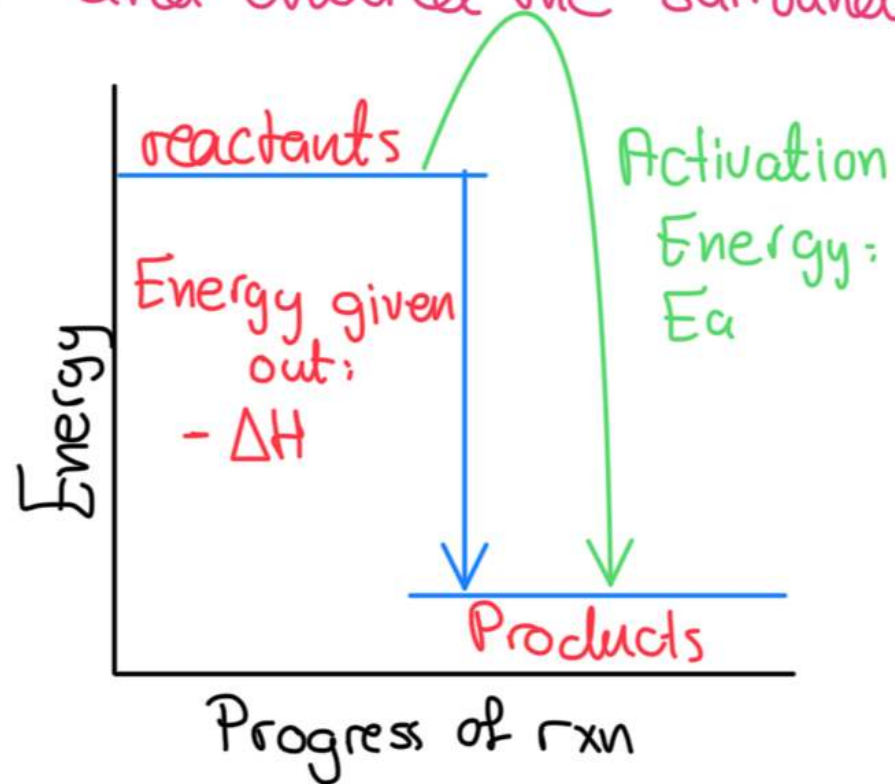
ΔH : Enthalpy change of the reaction: The transfer of thermal energy during a reaction.

① Exothermic reactions:

→ Products have lower energy than reactants



→ Exothermic reactions have negative ΔH , which shows that the energy content of the system has decreased, as some energy left the system and entered the surroundings.



* Temperature of surroundings increase:

(+)

* Energy of the system decrease:

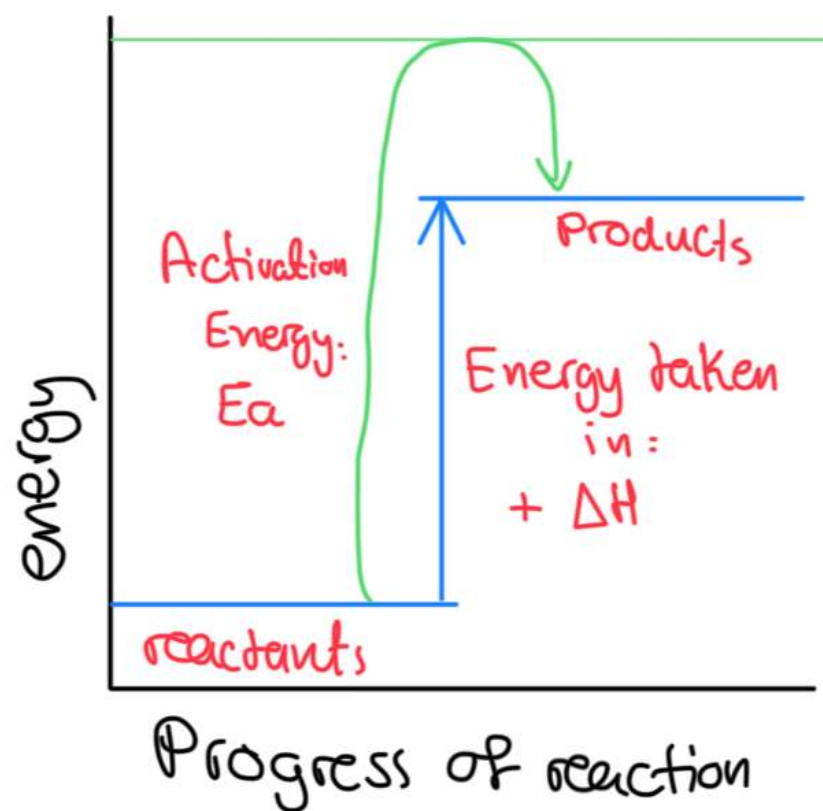
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② Endothermic reactions.

→ Products have higher energy than reactants as energy is taken from surroundings



→ Endothermic reactions have positive ΔH , which shows that the energy content of the system has increased, as it has gained energy from the surroundings.



* Temperature of the surroundings decrease:

(-)

* Energy of the system increase:

(+)

Breaking bonds $\longrightarrow$ Energy is taken in $\longrightarrow$ Endothermic

Forming bonds $\longrightarrow$ Energy is given out $\longrightarrow$ Exothermic

* At the start of a reaction, bonds are broken, so energy is taken in, so it is an endothermic reaction. Then, new bonds are formed, so energy is given out, so it is an exothermic reaction.

Bond energy: The amount of energy in kilojoules associated with the breaking or making of one mole of chemical bonds in a molecular compound.

In exothermic reactions: the energy released by bond formation in the products is greater than the energy needed to break the bonds in reactants. So products store less energy than reactants.

In endothermic reactions: The energy required to break the bonds in the reactants is greater than the energy released when new bonds are formed in products. So, the total energy given out is less than the total energy taken in.

$$\Delta H = \text{Energy needed to break bonds} - \text{Energy needed to form bonds}$$

Fuel: Any substance that is burnt to provide energy.

Types of Fuels:

Fossil Fuels: Coal, petroleum (oil), natural gas (methane).

* Fossil Fuels are burnt in power stations, to heat water and get steam. The steam drives the turbines that generate electricity.

What makes a good Fuel?

- * Should be cheap
- * Available in large quantities
- * Safe to store and transport.
- * Easy to ignite and burn, causing no pollution.
- * Capable of releasing large amounts of energy

Problems with Fossil Fuels:

- Non-Renewable

- Pollution:

- * Carbon dioxide $\rightarrow$ causes global warming
- * Carbon monoxide $\rightarrow$ forms through incomplete combustion of carbon containing fuels. $\rightarrow$ Poisonous, reacts with hemoglobin in blood, and prevents it from carrying oxygen around the body
- * Sulphur dioxide $\rightarrow$ forms when coal and sulfur containing compounds are burned. $\rightarrow$ Causes respiratory problems, and irritates the eye and throat. Dissolves in rain to form acid rain, which attacks stonework in buildings, lowers the pH in rivers, and kills the marine life.

* Biofuels / bioethanol: An alcohol, C_2H_5OH . Formed from any plant material by fermentation.

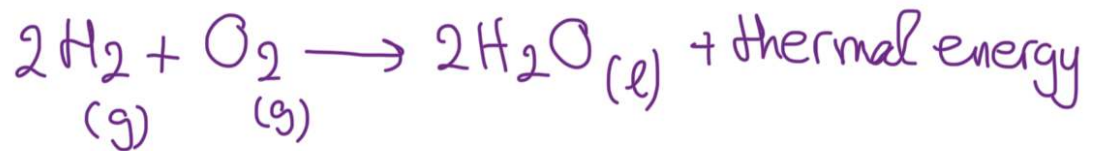
Advantages:

- * Made of renewable resources
- * More carbon neutral than fossil fuels. (The amount of carbon dioxide formed in the combustion is the same as the amount absorbed during tree's growth, then the wood is described as carbon neutral)

Disadvantages:

- * A lot of land is needed
- * Low yield.

* The hydrogen-oxygen cell (fuel cell): Hydrogen is a non-polluting fuel. It burns with oxygen to give out a large amount of thermal energy, and water is the only product formed.



1-The cell has two carbon electrodes coated with a catalyst, usually platinum, with a proton exchange membrane between them.

2-Hydrogen gives up electrons to the anode, forming H^+ ions, by oxidation.



3- Electrons flow to the cathode, where oxygen reacts with the electrons and the H^+ ions to form water, by reduction.



4-Water will be formed in the form of vapour

Advantages:

- * Water is the only product formed, which is harmless.
- * Renewable: Hydrogen is formed by electrolysis of water, which is plentiful, on adding a little acid.
- * They produce more energy per gram of fuel than other fuels
- * Operate with high efficiency

Disadvantages:

- * Storage of hydrogen is difficult, as hydrogen is a gas, and it must be stored as liquid under high pressure.
- * High likelihood of hydrogen leakage
- * Hydrogen filling stations are not widely available
- * Energy used to produce hydrogen and pressurizing the hydrogen comes from power plants that burn fossil fuels, which is linked to pollution and global warming.
- * Cars with hydrogen-oxygen fuel cells cost more than cars with petrol cars

Advantages of petrol engines:

- * Petrol filling stations are widely available for drivers.
- * Petrol is relatively cheap

Disadvantages:

- * Burning of petrol in car engines produces harmful substances:
 - Carbon dioxide, which is linked to climate change
 - Oxides of nitrogen, which is linked to acid rain & respiratory problems.
 - Deadly carbon monoxide from incomplete combustion
 - Unburnt volatile gases from the petrol.

* Nucleus fuels: They are not burned, they contain unstable atoms called radioisotopes. Overtime, they break down naturally into new atoms, giving out radiation and a lot of energy.

Nuclear reactors harness energy from the fission of uranium-235. The uranium is bombarded with speed neutrons, the collisions cause the nucleus of uranium-235 to split; which produces a large amount of energy is released. More neutrons are produced as well as atoms of lower mass such as krypton and barium. The new neutrons then hit other uranium-235 atoms & split them.

→ If the chain reaction is not stopped, it can cause an explosion.

Advantages:

- * No pollutants are produced
- * large amount of energy is produced.

Disadvantages:

- * The reactor produces highly radioactive waste materials. They are difficult to store and cannot be disposed easily.
- * leakage of radioactive materials have happened before.